Supplementary Material: Conchordal: Emergent Harmony via Direct Cognitive Coupling in a Psychoacoustic Landscape

Koichi Takahashi[1,2]
[1]Institute for Advanced Biosciences and Graduate School of Media and Governance, Keio University
[2]Conchordal.org
info@conchordal.org

S1. Algorithm Pseudocode

S1.1 Pitch Adaptation (Experiment 1)

Input: agents $\{a_1, \ldots, a_N\}$ with pitches x_i in $\log_2$-freq space.
Parameters: step size $\lambda = 15\,\text{ct}$, crowding weight $\lambda_C = 0.15$, $T_0 = 0.05$, $\tau = 30$.

`for` sweep $t = 0$ `to` 49:
  $T_t \leftarrow T_0 \cdot \exp(-t/\tau)$
  `for` each agent a_i:
    $\delta \sim \text{Uniform}(-\lambda, +\lambda)$; $x' \leftarrow x_i + \delta$
    $P(x) = \lambda_C \sum_{j \neq i} K_{\text{crowd}}(\Delta_{\text{erb}}(x, x_j))$
    $S_{\text{old}} \leftarrow s(t) \cdot C_{\text{field}}(x_i) - P(x_i)$
    $S_{\text{new}} \leftarrow s(t) \cdot C_{\text{field}}(x') - P(x')$
    `if` $U(0,1) < \min\big(1, \exp\big((S_{\text{new}} - S_{\text{old}})/T_t\big)\big)$ `then` $x_i \leftarrow x'$

where $s(t) = -1$ for $t \leq 24$ (dissonance-seeking), $+1$ for $t \geq 25$ (consonance-seeking).

S1.2 Metabolic Policy (Experiment 2)

Parameters: $c_b = 0.5$/s, $r_E = 0.4$/s, $\Delta t = 512/48000\,\text{s}$.

`for` each time step:
  `for` each agent a_i:
    $C_{\text{score}} \leftarrow C_{\text{field}}(x_i)$
    $E_i \leftarrow E_i - c_b \cdot \Delta t$ *(basal drain)*
    $E_i \leftarrow E_i + r_E \cdot \max(0, C_{\text{score}}) \cdot \Delta t$ *(continuous recharge)*
    `if` $E_i \leq 0$: agent dies; respawn at random pitch within ± 1 oct

S1.3 Shared Rhythm Scaffold (Experiment 3)

Parameters: $k_{\text{time}} = 3.0$, kick at 2 Hz, $\omega_0 = 1.8 \cdot 2\pi$ rad/s, $\pm 2\%$ jitter.

`for` each time step dt:
  Advance canonical beat phase $\theta(t)$ at 2 Hz
  `if` condition = scrambled `and` cycle boundary crossed: sample new offset $\psi \sim \text{Uniform}(0, 2\pi)$
  `for` each agent a_i:
    Set drive phase $\theta_i^* \leftarrow \theta(t)$ (shared), $\theta(t) + \psi$ (scrambled), or none (off)
    $k_{\text{eff},i} \leftarrow k_{\text{time}}$ (shared/scrambled) or 0 (off)
    $\dot{\phi}_i \leftarrow \omega_i + (k_{\text{eff},i}/N) \sum_j \sin(\phi_j - \phi_i) + F(\theta_i^*)$
    $\phi_i \leftarrow \phi_i + \dot{\phi}_i \cdot dt$
  Record PLV and onset phase relative to the canonical beat $\theta(t)$

S1.4 Hereditary Respawn (Hereditary Adaptation)

Key parameters: $N = 16$, range ± 2 oct, candidate count $K = 16$, slot-local settling $= 2$ ticks, family-local azimuth search radius 0.5 st.

`for` each time step:
  Advance metabolism using the polyphonic leave-one-out score (C_{score} with crowding/capacity penalties)
  `if` selection = on: recharge and parent-choice weights depend on current score
  `if` selection = off: use neutral turnover with a small background-death floor
  `if` agent a_i dies:
    `if` respawn = heredity:
      Choose parent a_p from survivors
      Form a parent-biased set of harmonic-family peaks under the shared scene filter
    `else` (matched random control):
      Sample K log-random candidates across tessitura and score them with the same scene filter
    Choose one target family / slot
    Search locally within the chosen family to set the final azimuth
    Respawn with initial energy E_0

Matched random control: adult pitch remains locked, the same metabolic policy is used, and the same short slot-local settling phase is applied; only the parent-biased family choice is removed.

S2. Normalization of H_{01} and R_{01}

Both harmonicity and roughness are normalized to $[0, 1]$ before entering the consonance core.

Harmonicity (H_{01})

The raw harmonicity potential from sibling projection (Eqs. 1–2 in the main text) is normalized by dividing by the self-harmonicity of a single harmonic tone at the reference position (which equals the kernel's maximum response):

$$H_{01}(x) = \mathrm{clamp}\left(\frac{H_{\mathrm{raw}}(x)}{H_{\mathrm{ref,max}}},\ 0,\ 1\right), \tag{S1}$$

where $H_{\mathrm{ref,max}} = 1.0$ (the kernel is pre-normalized so that a single harmonic stack produces unit peak).

Roughness (R_{01})

Raw roughness is first expressed as a ratio $\rho = R_{\mathrm{raw}}/R_{\mathrm{ref}}$, where R_{ref} is the roughness of a single-tone reference density (computed once at initialization). This ratio is then mapped through a piecewise saturation function with physiological constant k ($k = 1.0$ in all experiments):

$$R_{01}(\rho) = \begin{cases} 0 & \rho \le 0, \\ \dfrac{\rho}{1+k} & 0 < \rho < 1, \\ 1 - \dfrac{k}{\rho + k} & \rho \ge 1, \end{cases} \tag{S2}$$

clamped to $[0, 1]$. For $\rho < 1$ (below-reference roughness) the mapping is linear; for $\rho \ge 1$ it is a hyperbolic saturation that approaches 1 asymptotically as $\rho \to \infty$. At $\rho = 1$ the function is continuous: $R_{01}(1) = 1/(1+k)$. With $k = 1$, the reference roughness maps to $R_{01} = 0.5$.

S3. Complete Parameter Table

Table S1 lists all parameters used across experiments.

S3.1 Supplementary Mechanism Check: Configurable Spectral–Temporal Bridge

The relation between Conchordal's spectral and temporal axes is not treated as a uniquely determined neurocognitive law. Existing neuroscience motivates several plausible couplings between harmonic structure, salience, attention, and rhythmic prediction, but does not reduce them to a single fixed mapping. Conchordal therefore treats the bridge from spectral state to temporal coupling as a design degree of freedom available to the composer. The current implementation includes one such mapping: consonance modulates metabolic vitality, and vitality gates the effective coupling strength of the phase oscillator. In the supplementary assay shown here, the bridge uses $R = 1.0$/s, $\gamma = 0.5$/s, $E_{\mathrm{cap}} = 1.0$, $\lambda_v = 1.0$, and $v_{\mathrm{floor}} = 0.0$. Figure S1 reports this assay as an implementation-level mechanism check, not as evidence that consonance uniquely determines rhythmic coupling in biological cognition.

S4. Hereditary Adaptation: Current Supplementary Baselines

The current paper-facing hereditary assay is no longer the legacy parameter sweep reported in earlier drafts. The main text uses one fixed design and compares heredity against a *matched log-random* baseline that shares the same scene filter and short slot-local settling phase. Table S2 reports the current selection-on comparisons for the paper assay and two stronger supplementary random baselines. These additional baselines are useful diagnostically, but they are not equivalent controls for musical polyphony: the unrestricted hard-random restart can maximize raw C_{score} by collapsing into very few occupied bands.

The main hereditary claim is therefore intentionally conservative. It is made against the matched log-random baseline rather than against the unrestricted hard-random restart, because the latter is a much stronger optimizer and tends to over-concentrate into a few occupied bands. Under the matched baseline, the current assay supports the paper's mechanistic claim: lineage-biased family choice plus local azimuth search plus metabolic selection yields more musical, less diffuse polyphony than matched random respawn.

S5. Bilinear Consonance Core: Derivation

The consonance core (Eq. 5 in the main text) is the most general bilinear form for two inputs $H_{01}, R_{01} \in [0, 1]$:

$$C = a\,H_{01} + b\,R_{01} + c\,H_{01}\,R_{01} + d. \tag{S3}$$

$C_{\textbf{field}}$ $(a, b, c, d) = (1, -1.35, 1, 0)$:

$$C_{\mathrm{field}} = H_{01} - 1.35\,R_{01} + H_{01}\,R_{01} \tag{S4}$$

$$= H_{01}(1 + R_{01}) - 1.35\,R_{01}. \tag{S5}$$

When H_{01} is high (strong harmonicity), the $H_{01}\,R_{01}$ term partially compensates for roughness, reflecting the perceptual observation that harmonically rich intervals tolerate moderate beating. When $H_{01} = 0$, roughness is penalized at full strength.

$C_{\textbf{density}}$ $(a, b, c, d) = (1, 0, -1, 0)$:

$$C_{\mathrm{density}} = H_{01}(1 - R_{01}). \tag{S6}$$

This product is nonnegative and peaks where harmonicity is high and roughness is low. In Conchordal's interactive mode it serves as a spawn-position weight; it is not used in the experiments reported in this paper but is shown in Figure 1D for comparison.

$C_{\textbf{level01}}$: C_{field} passed through a logistic sigmoid $\sigma(\beta(x - \theta))$ with $\beta = 2$, $\theta = 0$. This bounds the metabolic signal to $[0, 1]$ while preserving the ranking.

S6. Statistical Status of the Current Hereditary Assay

Earlier versions of this project reported a 20-seed factorial heredity analysis with an exact sign-flip permutation test.

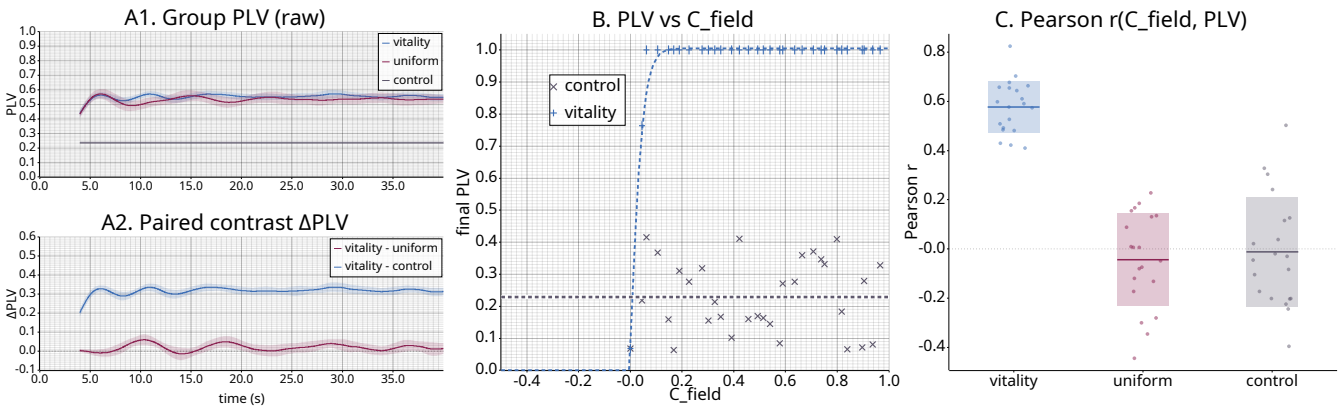

Figure S1: Supplementary mechanism check for one configurable spectral–temporal bridge (20 seeds × 3 conditions). (A1) Group mean PLV over time (95% CI). (A2) Paired contrast ΔPLV (95% CI). (B) Per-agent final PLV vs. C_{field} (representative seed) with sigmoid fit (c_{50} fixed at 0). (C) Seed-level Pearson $r(C_{\text{field}}, \text{PLV})$. In the current implementation, vitality-gated coupling yields a graded bridge from spectral state to entrainment strength; alternative bridge mappings remain possible within the broader Conchordal design.

That legacy test no longer matches the present paper-facing hereditary assay and has therefore been removed from the main argument. The current Figure 4 is a 4-seed mechanism rerun of the revised design, so the paper reports seed-level means, 95% confidence intervals, trajectory area (AUC_C), and threshold-crossing counts rather than reusing the obsolete legacy permutation result.

The intended next inferential step is a full 20-seed rerun of the *current* assay, after which a renewed nonparametric seed-level test can be reinstated. Until then, the hereditary result should be read as mechanism-level evidence rather than as the final word on cross-seed inferential robustness.

S7. Consonance-Only Control (Experiment 1)

Experiment 1 uses a two-phase curriculum: dissonance-seeking (steps 0–24) followed by consonance-seeking (steps 25–49). To assess whether the resulting polyphony depends on this schedule, we ran a *consonance-only* control with identical parameters but $s(t) = +1$ for all 50 sweeps (*local-search* condition, 20 seeds).

Both conditions reach indistinguishable final consonance ($C_{\text{score}} = 0.731$; $p = 0.98$), confirming that consonant structure emerges from the landscape–penalty interaction regardless of schedule. However, the curriculum significantly improves pitch diversity: more unique bins (9.4 vs. 5.9; $p < 0.0001$) and wider nearest-neighbour spacing (0.324 vs. 0.049 st; $p = 0.0001$), while interval entropy does not differ significantly ($p = 0.19$). The dissonance-seeking phase thus serves as an initialisation strategy that broadens the pitch distribution, preventing premature collapse to a small number of positions.

S7.1 Derived Metrics: LOO Score and Scene Consonance

Leave-One-Out (LOO) Score. For each agent i, the field spectrum is recomputed with agent i's harmonic contribution removed, yielding a leave-one-out field F_{-i}. The agent's LOO consonance is the consonance field evaluated at its own position x_i against the remaining agents:

$$C_{\text{score},i} = C_{\text{field}}(F_{-i},\, x_i). \tag{S7}$$

The population mean defines the primary outcome metric:

$$C_{\text{score}} = \frac{1}{N}\sum_{i=1}^{N} C_{\text{score},i}. \tag{S8}$$

Scene Consonance $G(F)$. Scene harmonicity H_{soc} and roughness R_{soc} quantify the *emergent* harmonic and roughness structure beyond what individual agents would produce in isolation. Let $H(F)$ and $R(F)$ denote harmonicity and roughness evaluated on the full scene spectrum F, and let $H(\{i\})$ and $R(\{i\})$ denote the same quantities for agent i presented alone. Then:

$$H_{\text{soc}} = H(F) - \frac{1}{N}\sum_{i=1}^{N} H(\{i\}), \tag{S9}$$

$$R_{\text{soc}} = R(F) - \frac{1}{N}\sum_{i=1}^{N} R(\{i\}). \tag{S10}$$

Scene consonance applies the same bilinear core (Eq. 5 in the main text):

$$G(F) = a\, H_{\text{soc}} + b\, R_{\text{soc}} + c\, H_{\text{soc}}\, R_{\text{soc}}. \tag{S11}$$

By subtracting the mean singleton contribution, $G(F)$ isolates the consonance arising from inter-agent spectral interactions rather than individual tonal quality.

S8. Terrain Control and Robustness Analysis

Three supplementary analyses test whether the self-organised polyphony reported in Experiment 1 depends on the specific structure of the psychoacoustic landscape rather than on incidental parameter choices.

S8.1 Shuffled-Landscape Control

We destroy spatial coherence by applying a fixed random permutation of the C_{score} bin indices, generated once per run and held fixed throughout all sweeps. Agents therefore move on a stationary landscape with the same *marginal* score distribution but no local gradient structure. Table S4 compares the shuffled condition with the standard *local-search* condition ($n = 20$ seeds each).

C_{score} (ref) denotes the consonance score of each final configuration evaluated on the *original* (unshuffled) landscape, enabling a fair cross-condition comparison. Without coherent spatial gradients, agents still spread across the register (similar unique-bin count and nearest-neighbour spacing) thanks to the crowding penalty, but the interval structure degrades sharply: entropy rises from 1.52 to 3.36 nats ($p < 0.0001$), and C_{score} drops from 0.731 to 0.459 ($p < 0.0001$). The coherent psychoacoustic landscape is therefore necessary for agents to self-organise into musically structured pitch configurations.

S8.2 Harmonicity–Roughness Misalignment Control

The consonance kernel combines harmonicity (H_{01}) and roughness (R_{01}). We test whether their spatial alignment matters by circularly shifting R_{01} by Δ cents before scoring, which preserves each component's marginal statistics but breaks the psychoacoustically motivated registration between them. Four shift values span the range $\Delta \in \{137, 271, 389, 523\}$ cents. Table S5 reports the results ($n = 20$ seeds per shift, 80 runs total for the misaligned aggregate).

All four misaligned conditions produce significantly fewer unique pitch bins than the local-search baseline (all $p < 10^{-4}$, Welch t), dropping from 9.4 to $\approx$6 bins—comparable to the consonance-only control (S7). Nearest-neighbour distances also decrease significantly (all $p \leq 0.0002$). Entropy effects are mixed: two shifts reach $p < 0.05$ while two do not, and the aggregate is significant ($p = 0.0002$). Misalignment thus preserves consonance (all $C_{\text{score}} \geq 0.70$) but eliminates the diversity benefit of the curriculum, reinforcing the conclusion from the shuffled-landscape control (S8.1) that a coherent landscape gradient is essential for structured polyphony.

S8.3 Consonance Kernel Coefficient Sensitivity

The consonance kernel $C = a\,H_{01} + b\,R_{01} + c\,H_{01}\,R_{01} + d$ has four coefficients; the default is $(a, b, c, d) = (1.0, -1.35, 1.0, 0.0)$. We sweep $b \in \{-2.0, -1.35, -0.7, 0.0\}$ and $c \in \{0.0, 0.5, 1.0, 1.5\}$ with $a = 1$, $d = 0$ fixed (16 conditions, 20 seeds each). Table S6 reports the results; because the sweep varies the kernel while holding everything else constant, it isolates the sensitivity of the consonance landscape to its parameterisation.

Unique-bin counts (5.8–6.5) and entropy (1.61–1.84 nats) are stable across the entire (b, c) grid, showing that the emergent pitch structure is not fragile to kernel parameterisation. C_{score} increases when the roughness penalty $|b|$ decreases or the interaction weight c increases (0.729–0.814), which is expected from the kernel definition. The overall level of unique bins ($\approx$6) matches the consonance-only control (S7), confirming that the crowding penalty—not the kernel coefficients—is the primary driver of the higher diversity (9.4 bins) seen in the full local-search condition.

S8.4 Terrain Control (Hereditary Adaptation)

The updated hereditary assay has *not* yet been rerun on shuffled terrain. The corresponding legacy table was removed because it described an earlier design that no longer matches the current paper-facing hereditary mechanism. The present hereditary claim should therefore be read as mechanism-level evidence within one well-characterized psychoacoustic terrain, exactly as stated in the main text.

S8.5 Topology Generalization (Hereditary Adaptation)

Likewise, the updated hereditary assay has not yet been rerun across the partial-count topology battery used for the polyphony assay. No topology-generalization table is therefore reported here. When this battery is rerun, it will use the current paper-facing hereditary mechanism: family-biased respawn, local azimuth search, metabolic selection, and the matched log-random baseline.

S8.6 Phase Coherence (Hereditary Adaptation)

Phase-coherence readouts from earlier hereditary builds are not carried forward into the present paper-facing assay. The current hereditary claim is intentionally spectral and lineage-based: it concerns where offspring are placed in pitch space and how that placement accumulates across generations. Any future reintroduction of rhythmic-order statistics for the hereditary assay should be treated as a separate supplementary extension rather than as part of the current main claim.

Table S1: Complete parameter listing for all experiments.

Parameter	**Symbol**	**Exp. 1**	**Exp. 2**	**Exp. 3**	**Exp. 4**
Anchor frequency	f_0	220 Hz	220 Hz	220 Hz	220 Hz
Population size	N	24	32	32	16
Seeds		20	20	20	4
Pitch range		± 2 oct	± 1 oct	± 1 oct	± 2 oct
Harmonicity / Roughness					
Harmonic order	N_H	16	16	16	16
Power-law exponent	ρ	0.4	0.4	0.4	0.4
Roughness saturation	k	1.0	1.0	1.0	1.0
Consonance core					
C_{field} coefficients	(a, b, c, d)	$(1, -1.35, 1, 0)$	$(1, -1.35, 1, 0)$	$(1, -1.35, 1, 0)$	$(1, -1.35, 1, 0)$
C_{density} coefficients[‡]	(a, b, c, d)	$(1, 0, -1, 0)$	—	—	—
Sigmoid (β, θ)		—	$(2, 0)$	—	$(2, 0)$
Pitch adaptation					
Step size	λ	0.15 st	—	—	—
Crowding weight	λ_C	0.15	—	—	0.005
Capacity penalty		—	—	—	radius 35 ct; 3 free voices; weight 0.07
Initial temperature	T_0	0.05	—	—	—
Annealing time constant	τ	30 steps	—	—	—
Duration		50 sweeps	—	—	—
Phase switch		step 25	—	—	—
Burn-in discarded		10 steps	—	—	—
Metabolism					
Basal cost	c_b	—	0.5/s	—	0.12/s
Recharge coefficient	r_E	—	0.4/s	—	0.20/s (selection-on)
Energy cap	E_{cap}	—	1.0	—	1.0
Time step	Δt	—	512/48k s	0.02 s	512/48k s
Background death		—	—	—	0.03/s (selection-off only)
Termination		—	200 deaths	40 s	12k steps
Temporal coupling					
Base coupling	k_{time}	—	—	3.0	—
External kick	f_{kick}	—	—	2 Hz	—
Intrinsic frequency	ω_0	—	—	$1.8 \cdot 2\pi$	—
Frequency jitter		—	—	$\pm 2\%$	—
Heredity					
Respawn candidates		—	—	—	matched $K = 16$ in both arms
Parent proposal		—	—	—	contextual-C family peaks
Parent selection		—	—	—	score/energy-weighted (sel. on), uniform (sel. off)
Juvenile settling		—	—	—	2 ticks, 0.20 st radius
Azimuth policy		—	—	—	local search ± 0.5 st at 0.05 st grid
JI cross-validation					
Proximity kernel	α	—	—	—	4.0×10^{-4} ct^{-2}
Ratio limit	p, q	—	—	—	≤ 8
Ratio weight		—	—	—	$1/(pq)$

[‡] Visualization only (Figure 1D); not used in experiment algorithms.

Table S2: Current hereditary-adaptation baselines (selection-on arms, 4 matched seeds). The main paper comparison is heredity+selection vs. matched log-random+selection.

Condition	Baseline type	C_{score}	Entropy	JI_{score}	Unique bins
Heredity+selection	main heredity assay	0.549	3.259	0.399	10.75
Random+selection	matched log-random	0.341	4.325	0.191	14.00
Random+selection	scene-peak matched random	0.346	4.137	0.288	14.00
Random+selection	unrestricted hard random-log	0.837	1.165	0.598	5.25

Table S3: Curriculum vs. consonance-only control (Experiment 1, *local-search* condition, $n = 20$ seeds).

Metric	**Curriculum**	**Cons.-only**	t
C_{score} (post)	0.731 ± 0.057	0.731 ± 0.041	-0.02
Unique pitch bins	9.4 ± 1.2	5.9 ± 0.6	-11.6^{***}
NN distance (st)	0.324 ± 0.245	0.049 ± 0.068	-4.8^{***}
Interval entropy	1.52 ± 0.36	1.65 ± 0.27	1.4

Table S4: *Local-search* vs. shuffled landscape (Experiment 1).

Metric	**Local-search**	**Shuffled**	t
Unique pitch bins	9.4 ± 1.2	9.6 ± 1.0	0.6
NN distance (st)	0.324 ± 0.245	0.328 ± 0.207	0.1
Interval entropy	1.52 ± 0.36	3.36 ± 0.27	18.6^{***}
C_{score} (ref)	0.731 ± 0.057	0.459 ± 0.042	-17.3^{***}

Table S5: *Local-search* vs. H/R-misaligned conditions (Experiment 1).

Condition	Unique bins	NN dist. (st)	Entropy
Local-search	9.4 ± 1.2	0.324 ± 0.245	1.52 ± 0.36
$\Delta = 137\,¢$	5.9 ± 1.0	0.047 ± 0.081	1.72 ± 0.30
$\Delta = 271\,¢$	6.4 ± 0.8	0.062 ± 0.078	1.83 ± 0.29
$\Delta = 389\,¢$	6.2 ± 1.0	0.065 ± 0.103	1.74 ± 0.27
$\Delta = 523\,¢$	5.8 ± 0.9	0.032 ± 0.054	1.57 ± 0.31
Misaligned (avg)	6.1 ± 1.0	0.052 ± 0.082	1.72 ± 0.31

Table S6: Coefficient sweep: polyphony metrics across 16 (b, c) settings.

b	c	Unique bins	Entropy	C_{score}
-2.00	0.00	6.2 ± 1.0	1.65 ± 0.26	0.738 ± 0.053
-2.00	0.50	5.8 ± 1.1	1.61 ± 0.33	0.729 ± 0.051
-2.00	1.00	6.2 ± 1.1	1.63 ± 0.26	0.746 ± 0.043
-2.00	1.50	6.1 ± 1.2	1.67 ± 0.29	0.751 ± 0.042
-1.35	0.00	6.2 ± 1.2	1.66 ± 0.28	0.736 ± 0.047
-1.35	0.50	6.2 ± 1.0	1.67 ± 0.32	0.733 ± 0.048
-1.35	1.00	6.4 ± 0.7	1.84 ± 0.29	0.734 ± 0.035
-1.35	1.50	5.9 ± 1.1	1.61 ± 0.33	0.760 ± 0.043
-0.70	0.00	6.5 ± 1.0	1.74 ± 0.28	0.739 ± 0.042
-0.70	0.50	6.1 ± 1.0	1.71 ± 0.31	0.761 ± 0.034
-0.70	1.00	5.8 ± 1.2	1.61 ± 0.33	0.768 ± 0.040
-0.70	1.50	6.2 ± 0.8	1.66 ± 0.22	0.788 ± 0.041
0.00	0.00	6.3 ± 1.2	1.68 ± 0.31	0.763 ± 0.034
0.00	0.50	5.9 ± 1.1	1.75 ± 0.28	0.760 ± 0.042
0.00	1.00	6.2 ± 0.8	1.79 ± 0.27	0.792 ± 0.034
0.00	1.50	5.8 ± 0.9	1.71 ± 0.23	0.814 ± 0.028